amazon Performance Business Analytics

A parameter-by-parameter reading of the dashboard, with insights, recommendations, and data caveats

1,351 products analyzed

9 main categories

4.09 average rating

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About Amazon and the data behind the dashboard

About Amazon

- Founded in 1994 by Jeff Bezos as an online bookstore
- Now a global marketplace with its own and third-party sellers
- Ratings and discounts are visible, high-impact signals on every product page

The dataset

- Product listings with ID, name, category, actual and discounted price, discount %, rating, and rating count
- 1,465 raw rows became 1,351 unique products after cleaning
- Prepared in BigQuery, visualized in Looker Studio

1 Raw data

1,465 rows uploaded to BigQuery

2 Cleansing

Duplicates removed, rating and discount fixed, category split

3 Dashboard

Looker Studio, 1,351 products

Problem statement and key questions

Problem statement

Without one consolidated view, it is hard to see which categories satisfy customers, whether discounts lift ratings, and which products draw the most engagement.

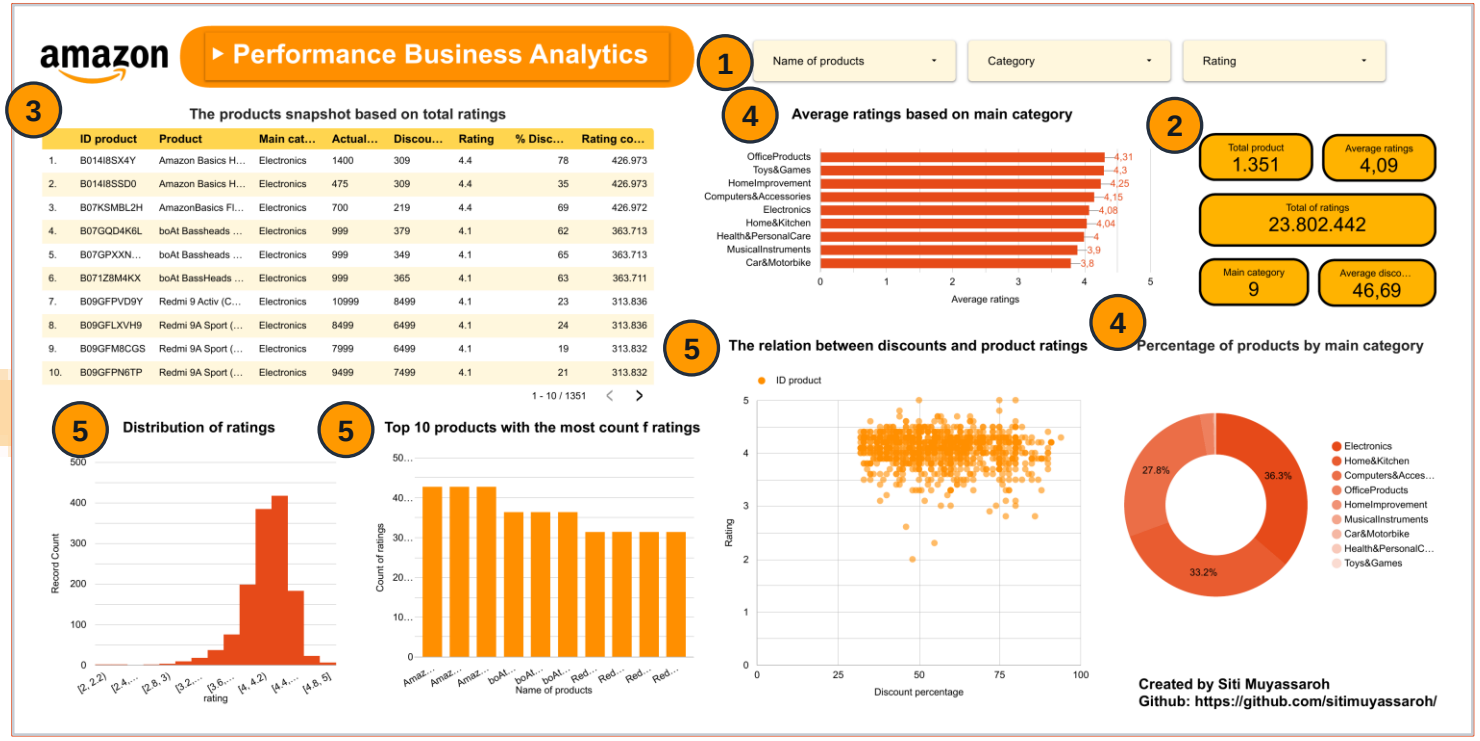
1. **How is the catalog spread across main categories?**
   Parameter: percentage of products by main category

2. **How satisfied are customers, overall and by category?**
   Parameters: average rating, rating distribution, average rating by category

3. **Do bigger discounts lead to better ratings?**
   Parameters: % discount, discount vs. rating scatter

4. **Which products attract the most ratings?**
   Parameters: rating count, top 10 chart, snapshot table

The dashboard at a glance



5 scorecards

1 snapshot table

5 charts

3 filters

1 Filters

Slice every chart by product name, category, or rating

2 Scorecards

Five headline KPIs for the whole catalog

3 Snapshot table

Top products by number of ratings, with price and discount

4 Category view

Average rating and share of products by main category

5 Ratings and discounts

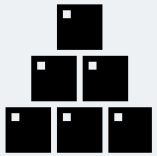
Rating spread, top 10 most-rated products, discount vs. rating

Scorecards: five headline numbers

1,351

Total products

Unique products after removing 114 duplicate rows from 1,465 raw rows.



4.09

Average rating

High on a 5-point scale. Category averages weighted by size reproduce it.



23.8M

Total of ratings

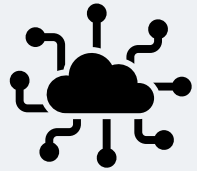
About 17,600 per product. Likely inflated because variants share counts.



9

Main categories

Three of them hold 97.3% of all products.



46.69%

Average discount

Close to half off list price, so discounting is the norm.



Read together: a large, concentrated, highly rated catalog where heavy discounting is the norm.

Top 10 products by number of ratings

#	Product ID	Product	Actual price	Discounted	Rating	% off	Rating count
1	B014I8SX4Y	Amazon Basics H…	1,400	309	4.4	78	**426,973**
2	B014I8SSD0	Amazon Basics H…	475	309	4.4	35	**426,973**
3	B07KSMBL2H	AmazonBasics Fl…	700	219	4.4	69	**426,972**
4	B07GQD4K6L	boAt Bassheads …	999	379	4.1	62	**363,713**
5	B07GPXXN…	boAt Bassheads …	999	349	4.1	65	**363,713**
6	B071Z8M4KX	boAt BassHeads …	999	365	4.1	63	**363,711**
7	B09GFPVD9Y	Redmi 9 Activ (C…	10,999	8,499	4.1	23	**313,836**
8	B09GFLXVH9	Redmi 9A Sport (…	8,499	6,499	4.1	24	**313,836**
9	B09GFM8CGS	Redmi 9A Sport (…	7,999	6,499	4.1	19	**313,832**
10	B09GFPN6TP	Redmi 9A Sport (…	9,499	7,499	4.1	21	**313,832**

All 10 rows are Electronics. Names and IDs are truncated as in the dashboard.

Only 3 product families

AmazonBasics, boAt, and Redmi fill all 10 slots.

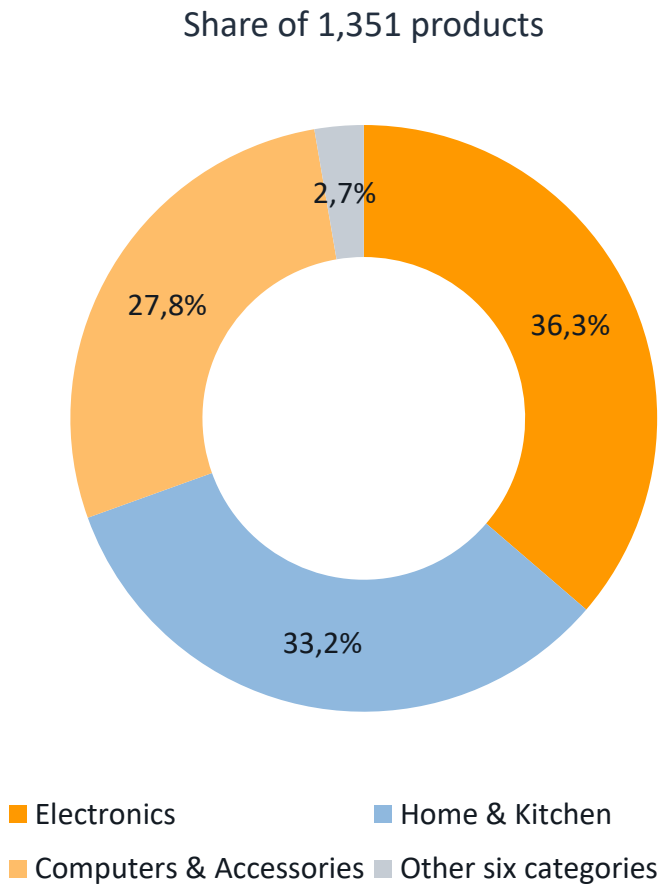
About 15% of all ratings

These 10 rows hold roughly 3.63M of the 23.8M ratings.

All rated 4.1 to 4.4

Popular products also score at or above the 4.09 average.

Product share by main category



Three categories dominate

Electronics (36.3%), Home & Kitchen (33.2%), and Computers & Accessories (27.8%) hold 97.3% of products.

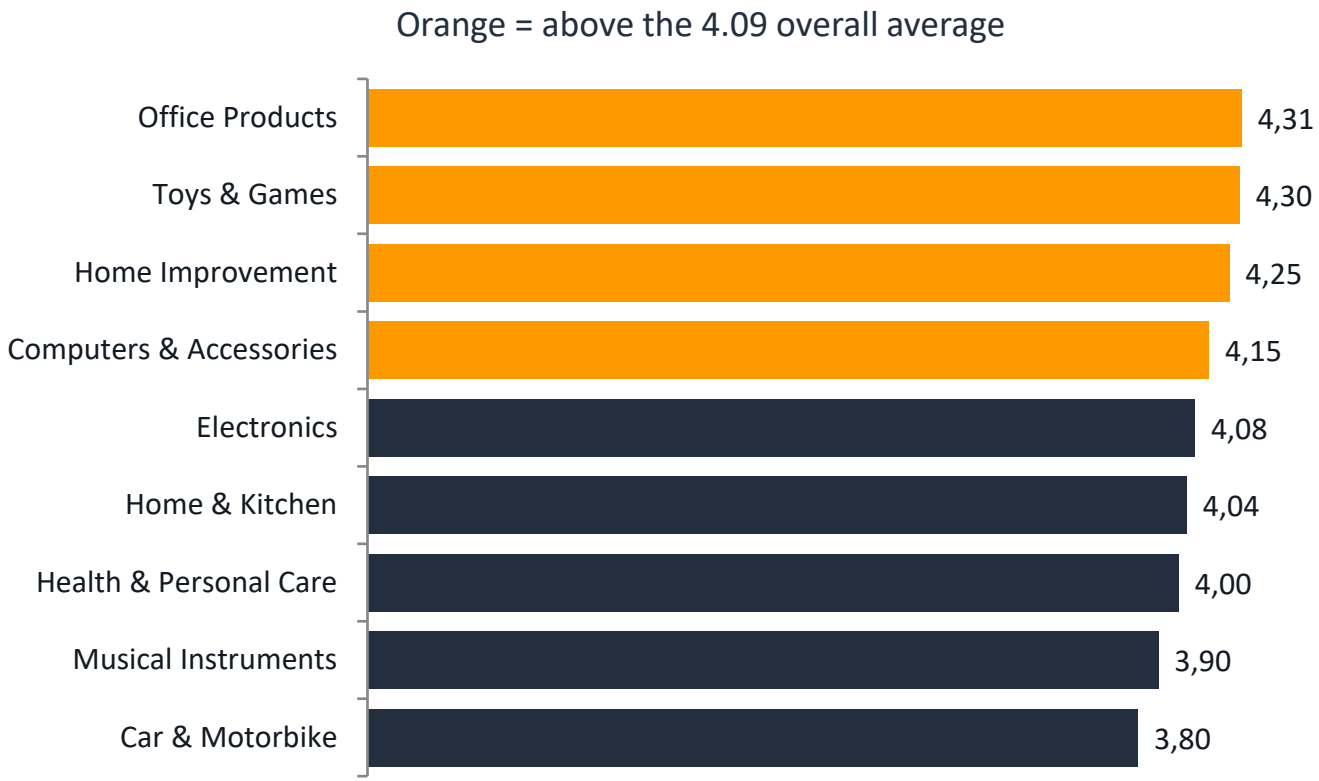
Six categories share 2.7%

The largest of the six is only about 2.3%. Several sit near 0.1% each, which is just one or two products.

Why it matters

Overall metrics mostly describe three categories, and averages for the small ones rest on very few products.

Average rating by main category



Check: weighting the three large categories by their share gives 4.09, matching the scorecard.

A narrow spread

Only 0.51 points separate the best (4.31) from the worst (3.80) category.

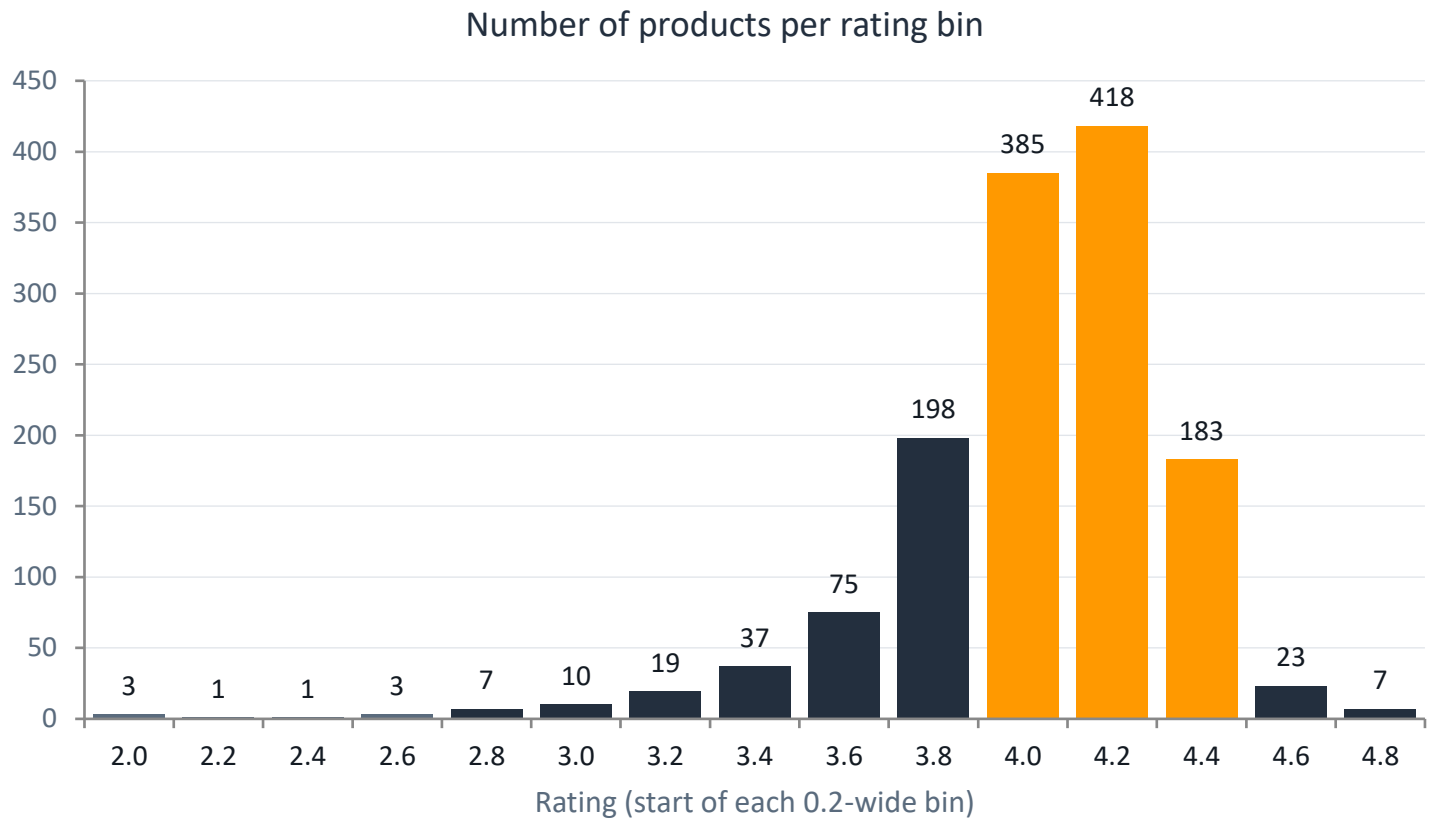
Big categories sit near average

Electronics (4.08) and Home & Kitchen (4.04) are just under 4.09. Computers & Accessories (4.15) is just over.

Small-sample caution

The top-rated categories are the smallest, so their scores are less reliable.

Distribution of ratings



Counts are approximate, read from the dashboard histogram. Orange = the 4.0 to 4.6 band.

Left-skewed, peaking at 4.2 to 4.4

About 420 products fall in the tallest bar.

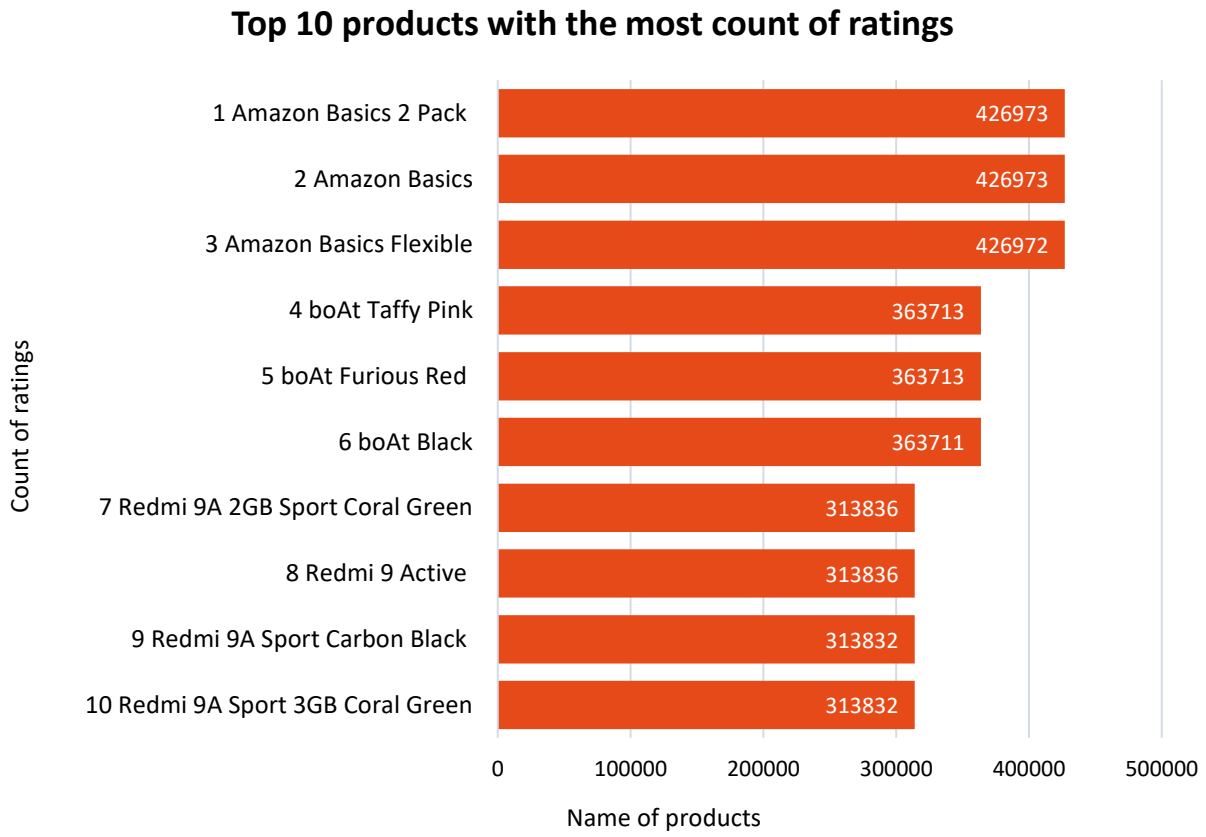
About 7 in 10 rate 4.0 to 4.6

A narrow, high band, so rating alone separates products poorly.

A thin tail below 3.6

Roughly 80 products (about 6%). These are the first candidates to review.

Products and count of ratings in the top 10



Orange = AmazonBasics, gray = boAt, blue = Redmi.

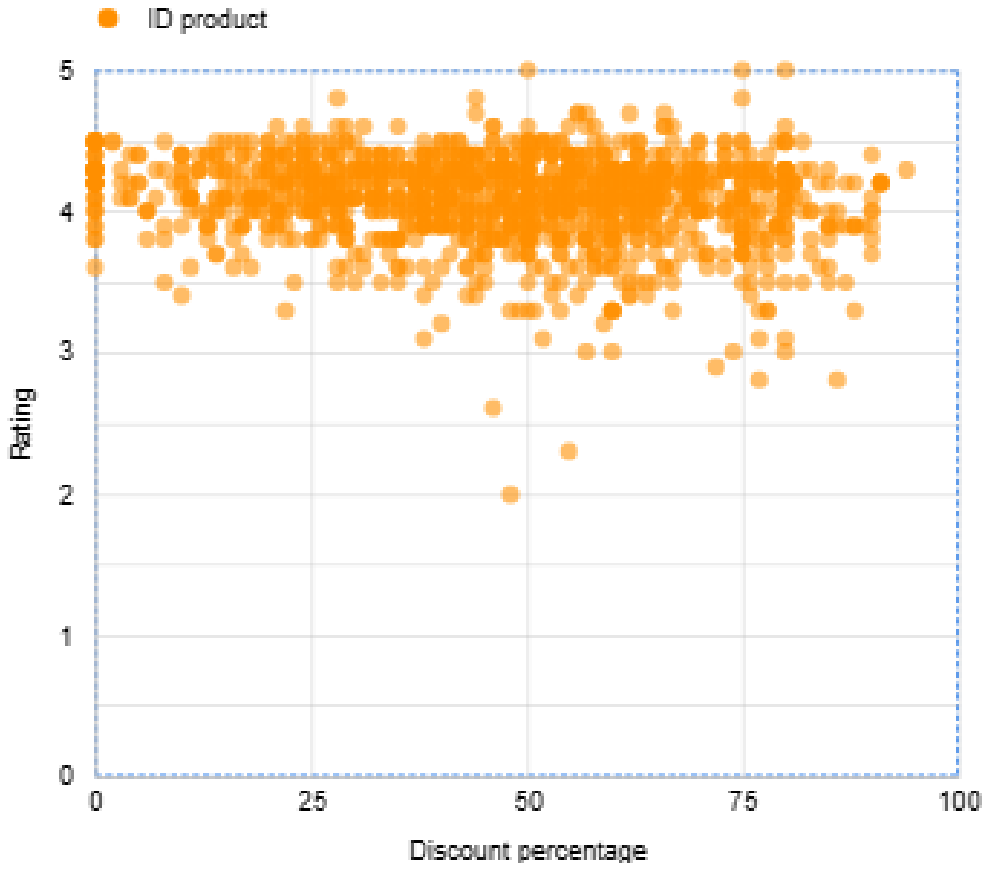
The top 10 is really only 3 product families. Amazon Basics (about 427K ratings), boAt (about 364K), and Redmi 9A/9 (about 314K) take all ten slots, and each family's variants have nearly identical counts.

The near-identical counts point to shared review pools, so the chart overstates breadth. The variants differ by only 1 to 4 ratings (for example 426,973 vs 426,972).

Rating volume shows popularity, not quality. The chart gives only the count of ratings, so these products are high-demand but not necessarily well-reviewed.

Discount versus rating

The relation between discounts and product ratings



1 **No visible slope**
The points form a flat cloud: heavy and light discounts earn similar ratings.

2 **Wide discount range**
Discounts run from roughly 30% to 94%, around the 46.69% average.

3 **Ratings cluster at 3.5 to 4.7**
A few outliers sit at 2.0 to 2.6 (about 42% to 55% off), and a handful reach 5.0.

What the parameters say together

1

Concentrated catalog

Three categories hold 97.3% of products, so overall metrics mostly describe them.

2

Consistently high ratings

The average is 4.09, and about 7 in 10 products sit between 4.0 and 4.6.

3

Ratings barely differentiate

Categories differ by only 0.51 points, so rating count and price add needed context.

4

Discounts are the norm

A 46.69% average discount, with no visible link to rating.

5

Popularity is concentrated

The top 10 are three product families and about 15% of all ratings.

6

Best scores, smallest samples

The highest-rated categories have the fewest products, so treat them as indicative.

Recommendations

NOW

Audit the low-rated tail

Filter Rating to products below 3.6 (roughly 6% of the catalog) and review quality and listing accuracy.

NOW

Fix the measurement

Group variants by product family and raise the scatter row limit so totals and plots are reliable.

NEXT

Broaden the assortment

Add listings beyond the top three categories, starting with Office Products, the largest of the small six.

NEXT

Anchor promotions

Use the AmazonBasics, boAt, and Redmi families as anchors for bundles and cross-selling.

NEXT

Rethink discounting

Judge deals by selling price, not list price, and test discount depth by category.

LATER

Add richer data

Bring in sales, time, and review text to link ratings and discounts to outcomes.

Thank you!

Explore the full dashboard and more projects:

- **Amazon Performance Business Analytics Dashboard**
- **GitHub: Siti Muyassaroh**